

CODY LEE

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Building end-to-end data and retrieval pipelines on messy, real data — ingestion, reconciliation, RAG — with measurement methodology rooted in forensic engineering.

EDUCATION

University of California, Riverside · B.S. Computer Science

Expected Jun 2026

Relevant coursework: Natural Language Processing, Artificial Intelligence, Data Science Ethics, Probability & Statistics, Operating Systems

PROJECTS

Polymarket Trading Bot Autopsy

Apr–Jun 2026

Published: 2-page reflective summary → github.com/clee12111/polymarket-autopsy/blob/main/reflective_2page.pdf

- Documented three measurement bug classes in a self-built trading pipeline that inflated paper trading results **~135×** and made the strategy actively misleading: **the bots my paper trading flagged as best performed worst live**.
- Built **3-layer LLM classification pipeline** (Haiku → Sonnet → Opus) classifying **2,409 wallets** into 12 strategy archetypes; **180 paper bots, 417,008 simulated trades**. Tiered-model design: **\$30–40 total cost vs ~\$150 if run on Opus alone**.
- 45-day solo project across Polymarket, Binance, and on-chain data; pipeline reconciled millions of cross-source transactions.
- Built and operated live execution environment with wallet auth, kill switches, and trade logging; deployed real capital across 6 architectures over 35 hours of live execution.
- Published 15-page technical autopsy and 2-page reflective summary covering the bugs, the methodology, and the infrastructure decisions worth keeping.

Aether — LLM Workflow Engine

Apr 2026

- Built hybrid RAG pipeline over financial compliance documents — **BM25 + all-MiniLM-L6-v2 dense embeddings** in ChromaDB, fused via **Reciprocal Rank Fusion**, then reranked with a **flashrank cross-encoder (ms-marco-MiniLM-L-12-v2)**; planner/executor/critic agent loop with audit trails.
- Retrieval precision: **96% on the eval suite**, perfectly reproducible across 5 runs; end-to-end correctness 87%; average cost **\$0.65 per run** via tiered model routing (Opus plans, Haiku critiques).
- Engine/domain separation: retrieval and agent layers are generic; domain knowledge enters through configuration, so the architecture generalizes beyond financial documents.

ChainTax — Crypto Tax Engine

Feb–Mar 2026

- Built cross-source data ingestion and reconciliation pipeline: **540K+ events in a single pipeline run** across Hyperliquid API and two on-chain APIs (Alchemy, DeBank), with bridge detection so the same dollar isn't double-counted across chains.
- FIFO lot matching with SpecID retrospective comparison; three competing IRS funding treatments (Section 163(d), basis-adjustment, Section 165(c)(2)) produce three Form 8949 variants per pipeline run, with Section 1092 offsetting-position detection across spot, perp, and correlated pairs.

Production RAG Stack Forensics

In Progress, 2026

- Building production RAG system on the industry-standard stack: **LangGraph** orchestration, **Pinecone** vector DB, **Langfuse** observability, and cross-provider inference (**Claude Sonnet 4.6, GPT-5.4, Gemini 3.1 Pro**) over the FastAPI documentation corpus.
- Custom **MCP server** exposes forensic instrumentation (eval results, retrieval latency, failure-mode breakdown, per-stage cost) as agent-queryable tools — integration with the open Model Context Protocol now adopted across the AI tooling ecosystem.
- Engineering record across three living documents: chronological journal of architectural decisions, 3–5 documented failure modes (5-section template), locked architecture snapshot. Extends polymarket autopsy methodology to production AI infrastructure.

TECHNICAL SKILLS

Languages	Python, TypeScript, SQL, Bash
Data	Cross-source ingestion and reconciliation, pandas, DuckDB, SQLite, async websocket pipelines, ETL design
RAG / LLM	Hybrid retrieval (dense + sparse), chunking, embedding, reranking, vector DBs, LLM evaluation, prompt engineering, Anthropic/OpenAI SDKs
Methodology	Forensic measurement, failure-mode analysis, statistical evaluation. Git, REST APIs, Streamlit, VPS deployment.